

THE VILLAGES, FL, USA

WMO: 722012

Lat: 28.950N

Lon: 81.967W

Elev: 89

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 02-18

WBAN: 92817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	36.7	18.3	13.9	45.1	23.2	17.6	47.8	19.8	64.8	18.1	63.9	2.9	0	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	93.2	77.2	91.4	76.8	90.7	76.6	82.2	85.5	81.1	85.2	80.0	84.7	6.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	164.0	83.0	80.5	159.0	82.6	79.0	151.1	81.8	46.2	86.1	44.6	84.3	43.5	86.1	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.7	13.0	DB	29.2	95.8	2.5	1.5	27.4	96.9	26.0	97.7	24.6	98.6	22.8	99.6	(4)
			WB	25.4	84.0	2.9	3.9	23.3	86.8	21.7	89.1	20.0	91.3	17.9	94.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	71.3	57.6	59.9	65.2	70.7	76.5	80.7	82.0	82.0	80.2	73.5	65.9	61.2	(6)
(7)		DBStd	10.63	9.05	8.38	7.26	5.73	4.24	2.80	2.17	2.29	2.32	5.98	6.99	8.90	(7)
(8)		HDD50	68	35	16	2	0	0	0	0	0	0	0	1	14	(8)
(9)		HDD65	815	260	180	90	20	0	0	0	0	0	13	72	180	(9)
(10)		CDD50	7856	270	294	472	620	823	921	993	991	905	729	477	361	(10)
(11)		CDD65	3129	30	38	96	190	358	471	528	526	455	277	98	62	(11)
(12)		CDH74	27252	179	272	750	1648	3347	4224	5097	4942	3865	1976	595	357	(12)
(13)		CDH80	10300	14	32	144	477	1381	1821	2211	2050	1475	562	91	42	(13)
(14)	Wind	WSAvg	4.8	5.1	5.6	5.8	5.8	5.6	4.6	3.5	3.8	4.3	4.6	4.5	4.8	(14)
(15)	Precipitation	PrecAvg	49.1	2.8	2.4	3.6	2.4	3.1	7.6	6.5	7.4	6.3	3.2	1.6	2.3	(15)
(16)		PrecMax	61.5	6.7	5.8	8.7	7.5	14.3	14.6	10.1	14.0	15.3	7.0	4.5	8.5	(16)
(17)		PrecMin	29.6	0.4	0.5	0.0	0.4	0.4	1.0	3.3	4.4	2.1	0.6	0.4	0.4	(17)
(18)		PrecStd	8.0	1.8	1.5	2.6	1.6	3.0	3.3	1.9	2.3	3.5	1.9	1.1	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.6	83.6	86.3	90.5	93.2	95.1	95.4	94.9	91.4	90.0	86.0	83.6	(19)
(20)		0.4%	MCWB	71.2	71.3	70.1	72.4	72.0	76.5	78.1	78.9	77.0	77.2	77.0	78.0	(20)
(21)		2%	DB	79.1	81.0	83.8	87.9	91.0	92.8	93.1	92.9	90.5	87.9	82.1	81.2	(21)
(22)		2%	MCWB	68.6	68.1	68.8	71.0	72.2	76.3	77.5	79.1	76.6	75.1	72.1	72.3	(22)
(23)		5%	DB	75.1	78.5	81.4	84.4	90.0	90.9	91.2	91.1	89.9	84.5	80.7	78.9	(23)
(24)		5%	MCWB	66.2	66.7	67.1	69.3	71.6	76.2	77.3	78.8	76.4	73.9	70.0	70.2	(24)
(25)		10%	DB	72.4	73.3	79.0	82.2	87.9	89.9	90.3	90.1	88.1	82.5	77.5	74.8	(25)
(26)		10%	MCWB	65.1	64.0	65.6	67.6	71.6	76.1	77.3	78.7	76.1	72.3	68.0	68.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	75.2	74.3	79.1	78.3	81.7	82.3	90.9	83.7	81.4	78.6	79.5	(27)
(28)		0.4%	MCDWB	78.6	79.0	79.6	81.6	84.6	86.2	86.0	90.9	86.7	84.5	82.7	81.8	(28)
(29)		2%	WB	71.8	71.4	72.4	74.6	76.5	80.5	81.1	83.2	81.3	78.9	75.5	75.4	(29)
(30)		2%	MCDWB	75.2	76.3	78.4	81.5	84.3	85.9	85.7	85.3	84.1	82.0	79.4	77.3	(30)
(31)		5%	WB	69.1	69.5	70.4	72.6	75.2	79.2	80.0	81.3	79.6	77.3	73.0	72.3	(31)
(32)		5%	MCDWB	73.7	74.8	76.7	80.2	83.4	85.3	85.4	85.6	83.1	81.8	76.9	76.0	(32)
(33)		10%	WB	66.5	66.9	68.7	71.0	73.9	78.1	79.0	80.0	78.4	75.6	71.0	69.9	(33)
(34)		10%	MCDWB	71.3	72.2	74.9	78.8	81.6	85.0	85.1	85.0	83.2	80.4	75.0	74.0	(34)
(35)	Mean Daily Temperature Range		MDBR	21.4	22.1	22.4	22.1	21.0	17.1	16.7	16.0	16.0	18.2	20.3	20.7	(35)
(36)		5% DB	MCDBR	21.8	23.4	23.1	23.1	22.7	19.0	18.1	17.5	17.4	17.9	19.9	20.2	(36)
(37)		5% DB	MCWBR	13.0	12.2	9.6	8.5	6.8	6.4	6.2	7.0	6.4	7.6	10.4	12.0	(37)
(38)		5% WB	MCDBR	19.8	20.3	19.7	19.8	19.0	16.0	16.2	15.7	14.9	14.6	16.3	18.3	(38)
(39)		5% WB	MCWBR	13.5	13.4	10.1	10.0	7.1	7.1	6.8	8.3	7.2	7.5	10.1	13.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.350	0.367	0.388	0.426	0.456	0.485	0.468	0.420	0.392	0.354	0.341	(40)	
(41)		taud	2.519	2.491	2.439	2.356	2.301	2.289	2.208	2.270	2.390	2.437	2.494	2.530	(41)	
(42)		Ebn at Noon	286	292	292	288	276	266	258	262	272	273	278	279	(42)	
(43)		Edh at Noon	29	32	36	40	42	43	46	43	37	33	29	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1069	1271	1629	1951	2050	1836	1780	1705	1543	1407	1153	978	(44)	
(45)		RadStd	77	101	135	114	132	148	116	115	106	102	63	82	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (44 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-11	-94	+286	+207	(47)

Nomenclature: See separate page